

alamp@Amps-MacBook-Air Homework_2 % ./hw2_cnn_torch.py

Number of trainable parameters: 394530

epoch 1 / 10:

[3200/48000]	loss: 0.664546,	accuracy: 71.8750%
[6400/48000]	loss: 0.328042,	accuracy: 90.6250%
[9600/48000]	loss: 0.472702,	accuracy: 87.5000%
[12800/48000]	loss: 0.437710,	accuracy: 84.3750%
[16000/48000]	loss: 0.410940,	accuracy: 84.3750%
[19200/48000]	loss: 0.486986,	accuracy: 78.1250%
[22400/48000]	loss: 0.374010,	accuracy: 90.6250%
[25600/48000]	loss: 0.330842,	accuracy: 87.5000%
[28800/48000]	loss: 0.223020,	accuracy: 93.7500%
[32000/48000]	loss: 0.334519,	accuracy: 87.5000%
[35200/48000]	loss: 0.298105,	accuracy: 93.7500%
[38400/48000]	loss: 0.382806,	accuracy: 84.3750%
[41600/48000]	loss: 0.296237,	accuracy: 90.6250%
[44800/48000]	loss: 0.257583,	accuracy: 90.6250%
[48000/48000]	loss: 0.412083,	accuracy: 87.5000%

TRAIN: Average loss: 0.470489, Accuracy: 39801/48000 (82.9188%)

VALID: Average loss: 0.368103, Accuracy: 10332/12000 (86.1000%)

epoch 2 / 10:

[3200/48000]	loss: 0.297969,	accuracy: 90.6250%
[6400/48000]	loss: 0.229695,	accuracy: 93.7500%
[9600/48000]	loss: 0.297550,	accuracy: 87.5000%
[12800/48000]	loss: 0.254323,	accuracy: 93.7500%
[16000/48000]	loss: 0.445488,	accuracy: 78.1250%
[19200/48000]	loss: 0.179563,	accuracy: 93.7500%
[22400/48000]	loss: 0.312519,	accuracy: 90.6250%
[25600/48000]	loss: 0.102070,	accuracy: 100.0000%
[28800/48000]	loss: 0.445957,	accuracy: 84.3750%
[32000/48000]	loss: 0.166802,	accuracy: 93.7500%
[35200/48000]	loss: 0.277530,	accuracy: 93.7500%
[38400/48000]	loss: 0.268610,	accuracy: 90.6250%
[41600/48000]	loss: 0.161065,	accuracy: 93.7500%
[44800/48000]	loss: 0.433411,	accuracy: 81.2500%
[48000/48000]	loss: 0.408644,	accuracy: 87.5000%

TRAIN: Average loss: 0.300622, Accuracy: 42757/48000 (89.0771%)

VALID: Average loss: 0.294508, Accuracy: 10692/12000 (89.1000%)

epoch 3 / 10:

[3200/48000]	loss: 0.171925,	accuracy: 90.6250%
[6400/48000]	loss: 0.313532,	accuracy: 84.3750%
[9600/48000]	loss: 0.285354,	accuracy: 90.6250%
[12800/48000]	loss: 0.235355,	accuracy: 87.5000%
[16000/48000]	loss: 0.204085,	accuracy: 93.7500%
[19200/48000]	loss: 0.069612,	accuracy: 96.8750%
[22400/48000]	loss: 0.311300,	accuracy: 93.7500%
[25600/48000]	loss: 0.044048,	accuracy: 100.0000%
[28800/48000]	loss: 0.189371,	accuracy: 90.6250%
[32000/48000]	loss: 0.364069,	accuracy: 93.7500%
[35200/48000]	loss: 0.114470,	accuracy: 93.7500%

[38400/48000] loss: 0.065385, accuracy: 100.0000%
[41600/48000] loss: 0.109359, accuracy: 93.7500%
[44800/48000] loss: 0.359231, accuracy: 84.3750%
[48000/48000] loss: 0.495591, accuracy: 87.5000%

TRAIN: Average loss: 0.254246, Accuracy: 43475/48000 (90.5729%)

VALID: Average loss: 0.271474, Accuracy: 10801/12000 (90.0083%)

epoch 4 / 10:

[3200/48000] loss: 0.144592, accuracy: 93.7500%
[6400/48000] loss: 0.093733, accuracy: 96.8750%
[9600/48000] loss: 0.070669, accuracy: 96.8750%
[12800/48000] loss: 0.359478, accuracy: 87.5000%
[16000/48000] loss: 0.186687, accuracy: 90.6250%
[19200/48000] loss: 0.307304, accuracy: 84.3750%
[22400/48000] loss: 0.189936, accuracy: 93.7500%
[25600/48000] loss: 0.097617, accuracy: 100.0000%
[28800/48000] loss: 0.050561, accuracy: 100.0000%
[32000/48000] loss: 0.222527, accuracy: 90.6250%
[35200/48000] loss: 0.237690, accuracy: 93.7500%
[38400/48000] loss: 0.344867, accuracy: 90.6250%
[41600/48000] loss: 0.633523, accuracy: 81.2500%
[44800/48000] loss: 0.158567, accuracy: 93.7500%
[48000/48000] loss: 0.191320, accuracy: 93.7500%

TRAIN: Average loss: 0.220953, Accuracy: 44030/48000 (91.7292%)

VALID: Average loss: 0.251987, Accuracy: 10900/12000 (90.8333%)

epoch 5 / 10:

[3200/48000] loss: 0.172888, accuracy: 93.7500%
[6400/48000] loss: 0.176271, accuracy: 96.8750%
[9600/48000] loss: 0.204962, accuracy: 93.7500%
[12800/48000] loss: 0.394808, accuracy: 87.5000%
[16000/48000] loss: 0.145612, accuracy: 96.8750%
[19200/48000] loss: 0.159449, accuracy: 93.7500%
[22400/48000] loss: 0.177323, accuracy: 93.7500%
[25600/48000] loss: 0.091863, accuracy: 96.8750%
[28800/48000] loss: 0.265450, accuracy: 93.7500%
[32000/48000] loss: 0.159828, accuracy: 90.6250%
[35200/48000] loss: 0.079801, accuracy: 100.0000%
[38400/48000] loss: 0.195051, accuracy: 93.7500%
[41600/48000] loss: 0.124132, accuracy: 93.7500%
[44800/48000] loss: 0.122858, accuracy: 96.8750%
[48000/48000] loss: 0.199210, accuracy: 90.6250%

TRAIN: Average loss: 0.190808, Accuracy: 44643/48000 (93.0062%)

VALID: Average loss: 0.236393, Accuracy: 10975/12000 (91.4583%)

epoch 6 / 10:

[3200/48000] loss: 0.081429, accuracy: 96.8750%
[6400/48000] loss: 0.075365, accuracy: 96.8750%
[9600/48000] loss: 0.142012, accuracy: 93.7500%
[12800/48000] loss: 0.367009, accuracy: 84.3750%
[16000/48000] loss: 0.199369, accuracy: 90.6250%
[19200/48000] loss: 0.213580, accuracy: 93.7500%
[22400/48000] loss: 0.268738, accuracy: 90.6250%

[25600/48000] loss: 0.275743, accuracy: 90.6250%
[28800/48000] loss: 0.052024, accuracy: 100.0000%
[32000/48000] loss: 0.503558, accuracy: 87.5000%
[35200/48000] loss: 0.145760, accuracy: 93.7500%
[38400/48000] loss: 0.267615, accuracy: 84.3750%
[41600/48000] loss: 0.238654, accuracy: 87.5000%
[44800/48000] loss: 0.100992, accuracy: 96.8750%
[48000/48000] loss: 0.249906, accuracy: 90.6250%
TRAIN: Average loss: 0.165301, Accuracy: 45059/48000 (93.8729%)
VALID: Average loss: 0.253429, Accuracy: 10949/12000 (91.2417%)

epoch 7 / 10:

[3200/48000] loss: 0.165765, accuracy: 96.8750%
[6400/48000] loss: 0.204237, accuracy: 90.6250%
[9600/48000] loss: 0.186245, accuracy: 93.7500%
[12800/48000] loss: 0.115773, accuracy: 96.8750%
[16000/48000] loss: 0.068705, accuracy: 96.8750%
[19200/48000] loss: 0.067615, accuracy: 100.0000%
[22400/48000] loss: 0.150726, accuracy: 93.7500%
[25600/48000] loss: 0.118182, accuracy: 93.7500%
[28800/48000] loss: 0.233834, accuracy: 96.8750%
[32000/48000] loss: 0.116944, accuracy: 96.8750%
[35200/48000] loss: 0.116133, accuracy: 93.7500%
[38400/48000] loss: 0.093757, accuracy: 93.7500%
[41600/48000] loss: 0.065200, accuracy: 96.8750%
[44800/48000] loss: 0.241554, accuracy: 87.5000%
[48000/48000] loss: 0.071664, accuracy: 96.8750%
TRAIN: Average loss: 0.143601, Accuracy: 45444/48000 (94.6750%)
VALID: Average loss: 0.256879, Accuracy: 10948/12000 (91.2333%)

epoch 8 / 10:

[3200/48000] loss: 0.036669, accuracy: 100.0000%
[6400/48000] loss: 0.121474, accuracy: 93.7500%
[9600/48000] loss: 0.154160, accuracy: 93.7500%
[12800/48000] loss: 0.070429, accuracy: 100.0000%
[16000/48000] loss: 0.071114, accuracy: 93.7500%
[19200/48000] loss: 0.133829, accuracy: 93.7500%
[22400/48000] loss: 0.032451, accuracy: 100.0000%
[25600/48000] loss: 0.085877, accuracy: 96.8750%
[28800/48000] loss: 0.074681, accuracy: 96.8750%
[32000/48000] loss: 0.042627, accuracy: 100.0000%
[35200/48000] loss: 0.121824, accuracy: 90.6250%
[38400/48000] loss: 0.088928, accuracy: 96.8750%
[41600/48000] loss: 0.204925, accuracy: 90.6250%
[44800/48000] loss: 0.325969, accuracy: 87.5000%
[48000/48000] loss: 0.052156, accuracy: 100.0000%
TRAIN: Average loss: 0.123898, Accuracy: 45790/48000 (95.3958%)
VALID: Average loss: 0.244851, Accuracy: 10979/12000 (91.4917%)

epoch 9 / 10:

[3200/48000] loss: 0.050935, accuracy: 96.8750%
[6400/48000] loss: 0.166742, accuracy: 93.7500%
[9600/48000] loss: 0.037454, accuracy: 100.0000%

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[12800/48000] loss: 0.172949, accuracy: 90.6250%
[16000/48000] loss: 0.120128, accuracy: 96.8750%
[19200/48000] loss: 0.177838, accuracy: 93.7500%
[22400/48000] loss: 0.079359, accuracy: 96.8750%
[25600/48000] loss: 0.169513, accuracy: 93.7500%
[28800/48000] loss: 0.136254, accuracy: 93.7500%
[32000/48000] loss: 0.228718, accuracy: 93.7500%
[35200/48000] loss: 0.217662, accuracy: 90.6250%
[38400/48000] loss: 0.155572, accuracy: 90.6250%
[41600/48000] loss: 0.045915, accuracy: 100.0000%
[44800/48000] loss: 0.054822, accuracy: 100.0000%
[48000/48000] loss: 0.066114, accuracy: 96.8750%
  TRAIN: Average loss: 0.106370, Accuracy: 46105/48000 (96.0521%)
  VALID: Average loss: 0.262777, Accuracy: 10992/12000 (91.6000%)
epoch 10 / 10:
[ 3200/48000] loss: 0.066740, accuracy: 100.0000%
[ 6400/48000] loss: 0.087691, accuracy: 96.8750%
[ 9600/48000] loss: 0.133983, accuracy: 93.7500%
[12800/48000] loss: 0.038295, accuracy: 100.0000%
[16000/48000] loss: 0.056558, accuracy: 100.0000%
[19200/48000] loss: 0.010233, accuracy: 100.0000%
[22400/48000] loss: 0.221061, accuracy: 93.7500%
[25600/48000] loss: 0.054583, accuracy: 96.8750%
[28800/48000] loss: 0.042138, accuracy: 96.8750%
[32000/48000] loss: 0.139685, accuracy: 93.7500%
[35200/48000] loss: 0.007908, accuracy: 100.0000%
[38400/48000] loss: 0.123870, accuracy: 93.7500%
[41600/48000] loss: 0.143330, accuracy: 93.7500%
[44800/48000] loss: 0.159006, accuracy: 93.7500%
[48000/48000] loss: 0.119585, accuracy: 93.7500%
  TRAIN: Average loss: 0.091319, Accuracy: 46366/48000 (96.5958%)
  VALID: Average loss: 0.308710, Accuracy: 10941/12000 (91.1750%)
  TEST: Average loss: 0.343697, Accuracy: 9059/10000 (90.5900%)
Accuracy on Test set is 90.5900%
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